

Distributed Computation for Marginal Likelihood based Model Choice

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Resumen basado en Buchholz, Ahfock & Richardson (2023) ¹.

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Motivation



- Computing the marginal likelihood in Bayesian model selection becomes challenging with massive datasets.
- Privacy, communication costs, and memory limitations may prevent centralization of the data.
- The authors propose a *divide-and-conquer* approach to estimate model evidence without sharing raw data between workers.

Hierarchical Representation of a Bayesian Model

- The Bayesian hierarchical structure can be expressed in reverse order over the data space $\mathcal{Y}$, the parameter spaces Θ_m , and the model space $\mathcal{M}$:

$$\begin{aligned}y \mid \theta, m &\sim p(y \mid \theta, m), & y \in \mathcal{Y}, \\ \theta \mid m &\sim p(\theta \mid m), & \theta \in \Theta_m, \\ m &\sim p(m), & m \in \mathcal{M}.\end{aligned}$$

Bayesian Modeling: Fundamental Quantities

- A Bayesian model specifies a likelihood $p(y \mid \theta, m)$ and a prior distribution $p(\theta \mid m)$ for parameters under model m .
- The posterior distribution updates prior beliefs after observing the data:

$$p(\theta \mid y, m) = \frac{p(y \mid \theta, m) p(\theta \mid m)}{p(y \mid m)} \propto p(y \mid \theta, m) p(\theta \mid m).$$

- The denominator,

$$p(y \mid m) = \int p(y \mid \theta, m) p(\theta \mid m) d\theta,$$

is the **marginal likelihood** or **model evidence**.

- This quantity measures how well model m explains the data, averaging over its parameter space.

Posterior Model Probability and Bayes Factor

- The posterior probability of model m given the data is

$$p(m | y) = \frac{p(y | m) p(m)}{\sum_{k=1}^K p(y | m_k) p(m_k)},$$

where $p(m)$ is the prior model probability.

- The **Bayes factor** comparing two models m_1 and m_2 is

$$BF_{12} = \frac{p(y | m_1)}{p(y | m_2)} = \frac{p(m_1 | y)}{p(m_2 | y)} \times \frac{p(m_2)}{p(m_1)}.$$

- It quantifies the evidence the data provide in favor of one model over another.
- Posterior model selection is driven jointly by model priors and Bayes factors.

Data Partitioning, Sub-Likelihoods, and Worker-Specific Posteriors

- The dataset is partitioned into S disjoint subsets:
 $y = (y_1, \dots, y_S), \quad y_s \in \mathcal{Y}_s.$
- Under conditional independence, the full likelihood factorizes as:

$$p(y \mid \theta, m) = \prod_{i=1}^n p(y_i \mid \theta, m) = \prod_{s=1}^S p(y_s \mid \theta, m).$$

- The posteriori distributioun given data shard y is:

$$p(\theta \mid y, m) \propto p(\theta \mid m) \prod_{s=1}^S p(y_s \mid \theta, m) \propto \prod_{s=1}^S p(\theta \mid m)^{1/S} p(y_s \mid \theta, m).$$

- The **subposterior** on worker s is thus

$$\tilde{p}(\theta \mid y_s, m) \propto p(y_s \mid \theta, m) p(\theta \mid m)^{1/S}.$$

- These subposteriors serve as the building blocks for reconstructing the full marginal likelihood.

Why the Model Posterior Does Not Factorize Across Workers

Cannot use a product-of-workers rule.

$$p(y) \not\propto \prod_{s=1}^S \tilde{p}(y_s)$$

$$p(m_i | y) \not\propto \prod_{s=1}^S \tilde{p}(m_i | y_s)$$

where, $\tilde{p}(m_i | y_s) \propto \tilde{p}(y_s | m_i)p(m_i)$

$$\tilde{p}(y_s | m_i) = \int_{\Theta} p(y_s | m_i, \theta) \tilde{p}(\theta, m_i) d\theta, \quad \tilde{p}(\theta | m_i) = \frac{p(\theta | m_i)^{1/S}}{\int_{\Theta} p(\theta | m_i)^{1/S} d\theta},$$

General Idea of the Approach

- Split the dataset into S non-overlapping subsets.
- Each worker computes a **subposterior** using a fractionated prior.
- The full marginal likelihood is recovered through a correction term that compensates for splitting the data.

Marginal

$$p(y) = \alpha^S \prod_{s=1}^S \tilde{p}(y_s) \int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta | y_s) d\theta$$

Model posterior

$$p(m_i | y) \propto \left(\prod_{s=1}^S \tilde{p}(m_i | y_s) \right) \alpha_i^S \int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta | y_s, m_i) d\theta$$

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$$p(y) = \alpha^S \prod_{s=1}^S \tilde{p}(y_s) \int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s) d\theta$$

Model posterior

$$p(m_i \mid y) \propto \left(\prod_{s=1}^S \tilde{p}(m_i \mid y_s) \right) \alpha_i^S \int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s, m_i) d\theta$$

- Unlike fixed-model inference, no simple product-of-experts rule exists for model selection.
- Subposterior distributions may systematically favor simpler models.
- The integral of the product of subposteriors must be approximated efficiently and accurately.

Method 1: Data Augmentation

Augmented conditional subposteriors. For each worker s , using latent variables z_s , the augmented subposterior satisfies

$$p(\theta \mid y, z) \propto p(\theta) p(y, z \mid \theta) \propto \prod_{s=1}^S p(y_s, z_s \mid \theta) p(\theta)^{1/S} \propto \prod_{s=1}^S \tilde{p}(\theta \mid y_s, z_s).$$

The conditional latent-variable subposteriors on each worker are

$$\tilde{p}(z_s \mid y_s, \theta) \propto p(y_s, z_s \mid \theta) \tilde{p}(\theta) \propto p(z_s \mid y_s, \theta) p(y_s \mid \theta),$$

so the subprior $\tilde{p}(\theta)$ is irrelevant when conditioning on θ .

Proposition 2. Using a data augmentation scheme for the augmented likelihood $p(y_{1:S}, z_{1:S} \mid \theta)$,

$$I_{\text{sub}} = \mathbb{E}_{\tilde{p}(z_{1:S} \mid y_{1:S})} \left[\int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s, z_s) d\theta \right],$$

where $\tilde{p}(z_{1:S} \mid y_{1:S}) = \prod_{s=1}^S \tilde{p}(z_s \mid y_s)$.

Method 1: Data Augmentation

- Introduce latent variables z_s on each shard to build a Gibbs sampler.
- Workers generate samples from $\tilde{p}(\theta, z_s \mid y_s)$.
- The subposterior integral is estimated via

$$\hat{I}_{\text{sub}} = \frac{1}{N} \sum_{i=1}^N \int_{\Theta} \prod_s \tilde{p}(\theta \mid y_s, z_s^{(i)}) d\theta.$$

Note: Applicable mainly to conditionally conjugate models. Where

$$\int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s, z_s^i) d\theta$$

is calculated analytically, and z_s^i are samples obtained from, e.g., a Gibbs sampler.

Algorithm 1: Distributed Model Evidence via Data Augmentation

Input: data y , number of chunks S , likelihood $p(\cdot \mid \theta)$, prior $p(\theta)$.

Split: Divide data into S chunks $y_1, \dots, y_S$.

Apply in parallel:

- 1: **for** $s = 1$ to S **do**
- 2: Sample $(\theta_s^1, z_s^1), \dots, (\theta_s^N, z_s^N) \sim \tilde{p}(\theta, z_s \mid y_s)$ using Gibbs sampling.
- 3: Calculate and store using Chib's method . $\tilde{p}(y_s) = \int p(y_s \mid \theta) \tilde{p}(\theta) d\theta$
- 4: **end for**

Combine:

$$p(y) = \alpha^S \prod_{s=1}^S \tilde{p}(y_s) \int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s) d\theta,$$

using

$$\int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s) d\theta = \mathbb{E}_{\tilde{p}(z_{1:S} \mid y_{1:S})} \left(\int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta \mid y_s, z_s) d\theta \right) \approx \hat{l}_{\text{sub}}.$$

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$$\int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta | y_s) d\theta = \mathbb{E}_{\tilde{p}(z_{1:S} | y_{1:S})} \left(\int_{\Theta} \prod_{s=1}^S \tilde{p}(\theta | y_s, z_s) d\theta \right) \approx \hat{l}_{\text{sub}}.$$

Method 2: Normal Approximation

- Each subposterior is approximated by a Gaussian:

$$\tilde{p}(\theta \mid y_s) \approx \mathcal{N}(\mu_s, \Sigma_s).$$

- Hence,

$$\begin{aligned} \int \prod_{s=1}^S \tilde{p}(\theta \mid y_s) d\theta &\approx \int_{\Theta} \prod_{s=1}^S \mathcal{N}(\theta \mid \mu_s, \Sigma_s) d\theta \\ &= \int_{\Theta} \mathcal{N}(\theta \mid \mu, \Sigma) d\theta \\ &= \exp\left(\sum_{s=1}^S \epsilon_s - \epsilon\right). \end{aligned}$$

- The product of Gaussian densities leads to a closed-form expression.

Error Bounds and Theoretical Guarantees

- The authors derive explicit bounds on the relative error of the approximation.
- The approximation error grows approximately linearly with the number of splits S .
- Accuracy improves when each shard contains a sufficiently large number of observations.

Extension: Distributed RJMCMC

- The modified product-of-experts rule enables model selection via RJMCMC in distributed environments.
- Workers estimate subposterior model probabilities, as well as $(\mu_{i,s}, \Sigma_{i,s})$ for each model.
- The global model probability satisfies

$$p(m_i | y) \propto \left[\prod_s \tilde{p}(m_i | y_s) \right] \alpha_i^S \int \prod_s N(\theta | \mu_{i,s}, \Sigma_{i,s}) d\theta.$$

Algorithm 3: RJMCMC Distributed Model Choice

Input: data y , number of chunks S , set of models m_k for $k = 1, \dots, K$, likelihoods $p(\cdot \mid \theta, m_k)$, priors $p(\theta \mid m_k)$, $p(m_k)$.

Split: Divide data into S chunks $y_1, \dots, y_S$.

Apply in parallel:

For $s = 1$ to S :

- Sample $\theta_s^1, \dots, \theta_s^{N_s} \mid m_k \sim \tilde{p}(\theta \mid y_s, m_k)$ using RJMCMC over all models k .
- Calculate and store $\tilde{p}(m_k \mid y_s)$, μ_s^k , and Σ_s^k for models $k = 1, \dots, K$.

Combine:

$$\hat{p}(m_k \mid y) \propto \left\{ \prod_{s=1}^S \tilde{p}(m_k \mid y_s) \right\} \alpha_k^S \left(\int_{\Theta} \prod_{s=1}^S \mathcal{N}(\theta \mid \mu_s^k, \Sigma_s^k) d\theta \right).$$

Bayes factors for any models (k, k') can be estimated as

$$BF_{k,k'} = \frac{\hat{p}(m_k \mid y) p(m_k)}{\hat{p}(m_{k'} \mid y) p(m_{k'})}.$$

Reversible Jump Markov Chain Monte Carlo (RJMCMC)

- **Current state:** model m of dimension d_m with parameters θ_m .
- **Move type:** choose either
 - *within-model move:* update θ_m , or
 - *between-model move:* propose a jump $m \rightarrow m'$.
- **Between-model proposal:** generate an auxiliary variable u and apply a reversible transformation

$$(\theta_m, u) \longrightarrow (\theta'_{m'}, u').$$

- **Acceptance probability:**

$$\alpha = \min \left\{ 1, \frac{p(y|\theta'_{m'}, m') p(\theta'_{m'}|m') p(m') q(m' \rightarrow m) q(u'|\theta'_{m'}, m' \rightarrow m) \left| \frac{\partial(\theta'_{m'}, u')}{\partial(\theta_m, u)} \right|}{p(y|\theta_m, m) p(\theta_m|m) p(m) q(m \rightarrow m') q(u|\theta_m, m \rightarrow m')} \right\}.$$

- **Accept or reject:** With probability α , move to $(m', \theta'_{m'})$; otherwise remain in (m, θ_m) .

Experiment 6: Distributed RJMCMC and Bayes Factor Stability

Experiment 6 examines the stability of Bayes factor estimation under the distributed RJMCMC sampler (Algorithm 3) as the dataset is split into an increasing number of subsets. A toy dataset of size $n = 4,000$ is generated with a binary outcome and five predictors, all with high pairwise correlation (0.9), creating a challenging environment for model selection. We evaluate the Bayes factors across several model comparisons:

Model 1/2 : reference log BF 3.2, Model 1/3 : reference log BF 3.6,
Model 2/3 : reference log BF 0.4.

Experiment 6: Distributed RJMCMC and Bayes Factor Stability

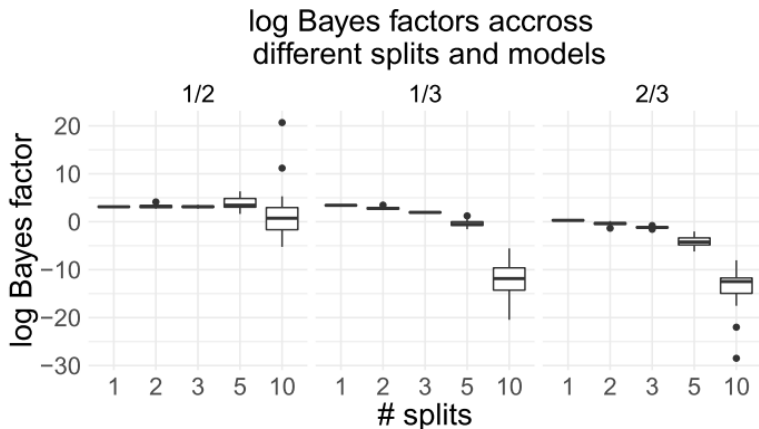


Figure: Comparison of the Bayes factor across multiple splits (y-axis) versus number of splits (x-axis).

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The results show that the distributed RJMCMC method preserves the reference Bayes factors even as the number of splits increases, highlighting its robustness and scalability.

- The normal approximation method is more stable and broadly applicable than the data-augmentation method.
- Experiments with very large datasets (Higgs, UK Biobank) show:
 - computation times reduced from weeks to hours,
 - small and stable bias across many splits,
 - consistent model ranking via Bayes factors.

- The proposed approach enables scalable Bayesian model choice without sharing raw data.
- The normal approximation strategy is efficient, general, and accurate for large datasets.
- Bayes factors and model selection decisions remain stable even under extensive partitioning.
- This framework is aligned with modern settings involving privacy constraints and federated learning.

Thanks

Questions? Comments?

Basic Identity (Chib, 1995)

For any point θ^* in the support of the posterior (typically the MAP or the posterior mean),

$$p(y) = \frac{p(y \mid \theta^*) p(\theta^*)}{p(\theta^* \mid y)}.$$

Taking logarithms,

$$\log p(y) = \log p(y \mid \theta^*) + \log p(\theta^*) - \log p(\theta^* \mid y).$$

The main difficulty lies in estimating the denominator $p(\theta^* \mid y)$, i.e., the value of the posterior density evaluated at θ^* . Chib (1995) proposes estimating this quantity using the output of an MCMC chain.

Estimating $p(\theta^* | y)$ under Gibbs Sampling

Assume the parameter vector is partitioned as

$$\theta = (\theta_1, \dots, \theta_K),$$

and the MCMC sampler is a Gibbs sampler with full conditionals $p(\theta_k | \theta_{-k}, y)$. Then Chib's identity implies

$$p(\theta^* | y) = p(\theta_1^* | y) p(\theta_2^* | \theta_1^*, y) \cdots p(\theta_K^* | \theta_1^*, \dots, \theta_{K-1}^*, y).$$

Each factor is estimated via Monte Carlo. For example,

$$\hat{p}(\theta_1^* | y) \approx \frac{1}{M} \sum_{m=1}^M p(\theta_1^* | \theta_{-1}^{(m)}, y),$$

where $\{\theta_{-1}^{(m)}\}_{m=1}^M$ are the posterior draws for all blocks except θ_1 .

Similarly,

$$\hat{p}(\theta_2^* | \theta_1^*, y) \approx \frac{1}{M} \sum_{m=1}^M p(\theta_2^* | \theta_{-2}^{(m)}, y),$$

using samples where θ_1 is fixed at θ_1^* and the remaining blocks vary according to the Gibbs structure.